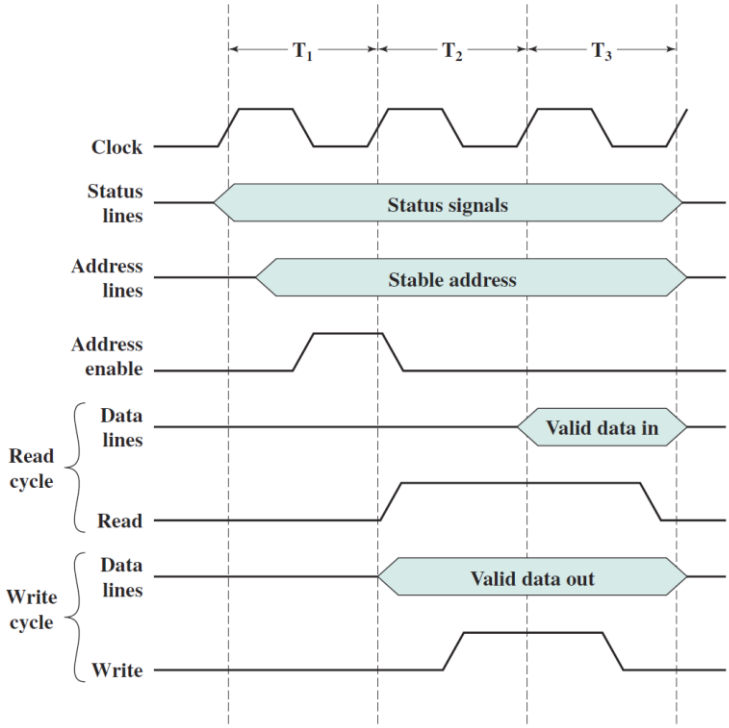
Computer Organization (CSD206): Tutorial 3

1. Consider a 32-bit microprocessor, with a 16-bit external data bus, driven by an 8-MHz input clock. Assume that this microprocessor has a bus cycle whose minimum duration equals four input clock cycles. What is the maximum data transfer rate across the bus that this microprocessor can sustain, in bytes/s?

2. Consider two microprocessors having 8- and 16-bit-wide external data buses, respectively. The two processors are identical otherwise and their bus cycles take just as long. Suppose all instructions and operands are two bytes long. By what factor do the maximum data transfer rates differ? Repeat assuming that half of the operands and instructions are one byte long.

3. For a synchronous read operation (see figure), the memory module must place the data on the bus sufficiently ahead of the falling edge of the Read signal to allow for signal settling. Assume a microprocessor bus is clocked at 10 MHz and that the Read signal begins to fall in the middle of the second half of T3.
   a) Determine the length of the memory read instruction cycle.
   b) When, at the latest, should memory data be placed on the bus? Allow 20 ns for the settling of data lines.



4. The Intel 8088 microprocessor has a read bus timing similar to that of Figure, but requires four processor clock cycles. The valid data is on the bus for an amount of time that extends into the fourth processor clock cycle. Assume a processor clock rate of 8 MHz
   a) What is the maximum data transfer rate?
   b) Repeat but assume the need to insert one wait state per byte transferred.

5. Consider a hypothetical 32-bit microprocessor having 32-bit instructions composed of two fields: the first byte contains the opcode and the remainder the immediate operand or an operand address.
   a) What is the maximum directly addressable memory capacity (in bytes)?
   b) Discuss the impact on the system speed if the microprocessor bus has:
      i. 32-bit local address bus and a 16-bit local data bus,
      ii. 16-bit local address bus and a 16-bit local data bus.

c) How many bits are needed for the program counter and the instruction register?